chapter ten

ESTIMATION AND HYPOTHESIS TESTING: TWO POPULATIONS

Knowledge is of two kinds.
We know a subject ourselves or
we know where we can find information upon it.
– Dr. Johnson

INTRODUCTION

In earlier chapters on hypothesis testing we used the value of a sample statistic to challenge an accepted value of a population parameter. We focused on both the mean and proportion. The goal was to show that there was a large enough difference between the value of the sample statistic and the population parameter that the difference could not be due to chance alone. When the difference was large, we argued that the value of the population parameter had changed. This approach, however, did not provide us any information about what was responsible for the change in the population parameter. In this chapter we introduce the idea of comparing two populations and use independent and dependent samples to help us determine what is responsible for the change.

When we work with **independent samples**, the two groups are often referred to as control and experimental. The control group represents what we might call the accepted value of the population parameter. The experimental group is given some treatment that we believe will change the value of the population parameter. We state a null hypothesis that the treatment used on the experimental group is not effective; that the difference in the population parameters is zero. We then compare the sample means. If the difference between the means is large enough, we argue the treatment was responsible for the change.

Interestingly enough, variance becomes a focus in this chapter. When we compare two population means, we need to be concerned about the population variances. If one of the variances is much larger than the other, the value of our test statistic may then be inflated causing us to make a type I error.

The work with independent samples is an effort to determine the effect of a treatment on the population parameter. Unfortunately the change may be a result of uncontrolled variables. One way of controlling confounding variables is to use **dependent samples** where data from the two samples is paired in a natural way. The test statistic used involves the differences of the paired data. The simplicity of this hypothesis test that the **mean difference** equals zero is complemented by the additional control of confounding variables.

In contrast, when we work with proportions, we are simply comparing two unknown population parameters. We seldom are worried about what may be responsible for the difference. We just want to know if there is a significant difference between population proportions. We state a null hypothesis that the two values are equal. If the difference between the two sample proportions is large enough we reject the null hypothesis.

CHAPTER 10 HOT SPOTS

1. **How Unequal Population Variances Can Cause Type I Errors.**
   Starts on **page 10–4.** Mann 10.1

2. **How To Decide If The Sample Data Comes From Dependent Samples.**
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3. **Why Is A Dependent Sample An Advantage In Hypothesis Testing?**
   Starts on **page 10–9.** Mann 10.3

4. **Deciding Which Test Statistic To Use For Independent Samples.**
   Starts on **page 10–13.** Mann 10.1, 10.2, 10.3

If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic for a help session with your professor.

✷
Mann Section 10.2

Hot Spot #1 – How Unequal Population Variances Can Cause Type I Errors

When we do a hypothesis test about the means of independent samples with n_1 or $n_2 < 30$, we must check to see if the variances of the two populations are equal. If we reject the hypothesis that population variances are equal, we then test for the null hypothesis that $\mu_1 = \mu_2$ using the degrees of freedom given by the formula:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

This value is typically smaller than $n_1 + n_2 - 2$. The critical point is that the result is a more conservative hypothesis test of $\mu_1 = \mu_2$ because the critical value for the t test statistic gets larger as the degrees of freedom get smaller. The reason we need to be more conservative is illustrated in the following example.

We will consider two populations that have the same mean but very different variances, and hence different standard deviations. The graphs of the two populations are shown below.

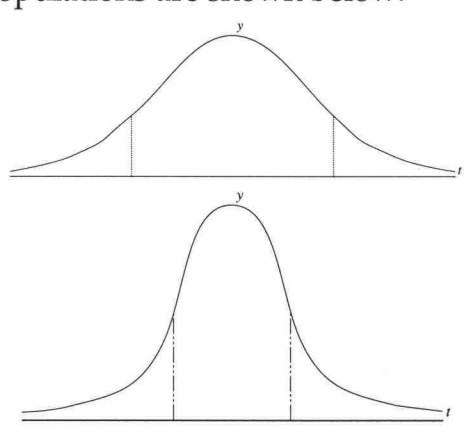

Figure 10–1: Two t Curves With The Same Mean And Differing Standard Deviations

Since the standard deviation is larger for the top curve (Figure 10–1), it is possible that we could get much of our data from one of the tails of the distribution. The smaller standard deviation for the bottom curve (Figure 10–1) makes it likely that most of our data will come from the center of the distribution. As such, the difference in sample means could be relatively large resulting in an incorrect decision that we reject the null hypothesis of equal population means. This would result in a type I error. The summary table below (Table 10–1) shows how this might occur. The table shows that there is a small difference in sample means, equal sample size, and a rather large difference in the sample variance.

Table 10–1: Summary Of Sample Sizes, Means, and Variances For Two t Curves

	n	$\bar{x}$	s^2
1	15	23.08	5.4275
2	15	21.90	1.7352

The standard error of $\bar{x}_1 - \bar{x}_2$ is $\sqrt{\frac{5.4275}{15} + \frac{1.7352}{15}} = 0.691$.

When we calculate the value of the t test, we get $t = \frac{23.08 - 21.90}{0.691} = 1.708$.

Now we need to look at the one-tailed critical value we would use if we *assume that the variances were equal*. The critical value would be $t(0.05, 28) = 1.701$ and *we would reject the null hypothesis*. However, if we *assume unequal population variances* and use the more conservative critical value for $df = 22$, we get $t(0.05, 22) = 1.717$ and *we fail to reject*. If the populations variances were unequal, the more conservative value of t using df for the unequal population variance test would keeps us from making a type I error. If we did get a t value larger than 1.717, we could be confident that we were rejecting the null hypothesis because there was a difference in population means.

Two different things were at work in the analysis we just finished. The first involves the effect of the variances on the numerator of the test

statistic. The second point relates to the effect of the variance on the denominator of the test statistic. Before we look in detail at these two points, it is important to emphasize that when we have unequal population variances we must be very cautious. This is true not only of our work with independent samples, but also of the work we will do later in what is known as analysis of variance (ANOVA). The following analysis assumes that the population means are actually equal.

Relative to the first point, we obtained a relatively low value of the sample mean from the first distribution (on the top of Figure 10–1) because of the larger standard deviation. If the standard deviations had been equal, we would not have found so many values on the low end of one of the curves. The resulting sample means would have been closer in value and the numerator of the test statistic would be smaller. The effect would be a smaller value of the test statistic. Of course this does not take into account what happens in the denominator if the variances are equal.

The second point that we need to consider is the effect of the unequal population variances on the denominator of the test statistic. If the distributions both had the larger variance, the denominator would be larger and further reduce the size of the resulting t value. This of course, would be coupled with the reduction in the value of the numerator as explained above. The result of the two effects would be an even smaller value of the test statistic. In any case we would fail to reject the null hypothesis, unless there actually was a difference in population means.

The following calculation shows what happens to the t value if the numerator does not change and we use equal sample variances using the larger value.

$$\text{The standard error of } \bar{x}_1 - \bar{x}_2 \text{ is } \sqrt{\frac{5.4275}{15} + \frac{5.4275}{15}} = 0.851.$$

When we calculate the value of the t test, we get $t = \dfrac{23.08 - 21.90}{0.851} = 1.387$.

The result is a smaller value of t. We would fail to reject the null hypothesis in both of the hypothesis tests we did. The large variance for both populations makes it likely to get sample means that differ by a large amount and we do not find enough evidence to reject the null hypothesis of equal population means.

If both distributions had the smaller variance, the denominator would become smaller. This would appear to increase the value of the test statistic. However, it would also be unlikely to get large differences in sample means when the variances are both small. Although it is more difficult to illustrate the effect in this case, if we did find a large value for the test statistic we would again be assured that it was a result of the treatment effect and not of a difference in population variance. The following calculation shows what happens to the t value if the numerator does not change and we use equal sample variances using the smaller value.

The standard error of $\bar{x}_1 - \bar{x}_2$ is $\sqrt{\dfrac{1.7352}{15} + \dfrac{1.7352}{15}} = 0.481$.

When we calculate the value of the t test, we get $t = \dfrac{23.08 - 21.90}{0.481} = 2.453$.

We would reject the null hypothesis in both of the hypothesis tests we did. The difference in means becomes more important with the smaller variances and results in a large t value. Our hypothesis was that the population means were equal making it highly unlikely that we would get a difference of means this large. So if we did get a difference this large, it would be very significant.

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Mann
Section 10.3

Hot Spot #2 – How To Decide If The Sample Data Comes From Dependent Samples

Statistical analysis is much simpler when we collect our own data because we will know if we used dependent samples. It is when we are asked to do a hypothesis test for data that we did not collect that we have to make a decision. We must then decide if the data is from an independent sample or a dependent sample. There are two situations that indicate that we have independent samples. The first is when the sample sizes are not equal. We then know that we must have independent samples because dependent samples require the same sample size for both data sets. The second situation is when we are given a summary table that includes the means, the standard deviations, and the sample sizes. It again is very clear that we have independent samples because we do not calculate individual means and standard deviations for dependent samples.

It is when the data itself is given for the two groups that have equal sample sizes that we run into trouble. We must then search for a key word or phrase to help us out. Obviously a key word is "dependent". It makes it easy when the problem includes the information that we have been given two dependent samples. One of the key phrases to look for is "paired differences". Another phrase that may be a key is that the data represents "before and after" scores on some measuring instrument such as a test. However, there are times when there may be reasonable doubt even after looking for key phrases. We may need to look carefully at the description of how the experiment was conducted.

Dependent samples often involve the same subjects or objects. We then obtain paired data values that reflect the changes as a result of the treatment. The idea involved in a dependent sample is that by using the same subject we do not introduce confounding variables. Unfortunately there are times when it is not possible to use the same subject. For

example, in studies about heart disease we often need to match patients as best we can for the confounding variables such as weight, age, life-style, etc. We then measure the treatment effect on the variable of interest, for example cholesterol or blood pressure. The paired data for the matched subjects is then treated as a dependent sample. The pairing of subjects makes it easier to find a treatment difference if it exists.

Naturally, this brings us to the point of asking, "what happens if we still can not decide if the data comes from a dependent sample?". The answer to the question is to use the independent sample hypothesis test. At worst we will make a type II error. If we instead make the assumption that the data reflects an effort to control confounding variables using the dependent sample, we could make a type I error. The possible errors are discussed further in the next **Hot Spot** that examines the effect of dependent samples on the hypothesis test.

Hot Spot #3 – Why Is A Dependent Sample An Advantage In Hypothesis Testing?

✹ *Mann Section 10.3*

The dependent sample gives us our first look at what is called blocking. We control confounding variables by matching subjects for the variables we want to control. The result is that we are more easily able to reject a false null hypothesis. The data set that follows (Table 10–2) illustrates the point. The example involves twenty-four males aged 25–29 selected from the Framingham Heart Study. Each of the twelve pairs of data represents a smoker and a non-smoker who were matched with regard to age and physical characteristics. The random variable measured was systolic blood pressure.

Table 10–2: Blood Pressures Of Twenty-Four Males In The Framingham Heart Study

Smokers	Non-Smokers
122	114
146	134
120	114
114	116
124	138
126	110
118	112
128	116
130	132
134	126
116	108
130	116

We are interested in testing the hypothesis that there is a difference in mean systolic blood pressure levels for the two populations. We will do the hypothesis test using the t test for dependent samples. We need to find the difference between each ordered pair. The differences are listed below:

Table 10–3: Ordered Pair Differences Of Blood Pressures In Framingham Heart Study

Smokers	Non-Smokers	d
122	114	8
146	134	12
120	114	6
114	116	-2
124	138	-14
126	110	16
118	112	6
128	116	12
130	132	-2
134	126	8
116	108	8
130	116	14

Using the differences as our random variable d, we find the following:

$$n = 12 \qquad \bar{d} = 6 \qquad s_d = 8.399134155$$

$$\frac{s_d}{\sqrt{12}} = 2.424621183$$

We are now ready to do the hypothesis test using the test statistic

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

Step 1. $H_0: \mu_S = \mu_{NS}$ versus $H_a: \mu_S \neq \mu_{NS}$.

Step 2. The α level will be used at 5%.

Step 3. The two-tailed critical value of t for $df = 12 - 1$ is ± 2.201.

Step 4. $t = \dfrac{6}{2.42462} = 2.47$

Step 5. Reject H_0.

Now we will see what happens if we did not know that the data involved paired data. The following summary table gives the necessary values for the hypothesis test. We will assume that the variances for the two populations are equal.

Table 10–4: Necessary Values For Hypothesis Testing Of Twenty-Four Males In Framingham Heart Study

	n	$\bar{x}$	s^2
Smokers	12	125.67	78.4242
Non-Smokers	12	119.67	102.4242

Technically, with the variances equal and a small sample size, we should use the t test that uses the pooled standard deviation. Since the sample sizes are equal, we can use the following test statistic that will give us the same value:

$$t = \frac{\bar{x}_S - \bar{x}_{NS}}{\sqrt{\frac{s_S^2}{n_S} + \frac{s_{NS}^2}{n_{NS}}}}$$

Step 1. $H_0: \mu_S = \mu_{NS}$ versus $H_a: \mu_S \neq \mu_{NS}$.

Step 2. The α level will be used at 5%.

Step 3. The two-tailed critical value of t for $df = 24 - 2$ is ± 2.074.

Step 4. $t = \dfrac{125.67 - 119.67}{\sqrt{\dfrac{78.4242}{12} + \dfrac{102.4242}{12}}} = 1.55$

Step 5. We fail to reject H_0.

Perhaps we need to review what we have just completed:

We first did a hypothesis test *assuming the data represented a dependent sample.* We found the t value of 2.47 in the critical region. So *we rejected the null hypothesis,* H_0.

We then did a hypothesis test *assuming the data represented an independent sample.* We found the t value of 1.55 that was **not** in the critical region. Thus, *we failed to reject the null hypothesis.*

The important point in all this is that when we treated the data as a dependent sample, we found a significant difference. When we assumed the data came from an independent sample we did **not** find a significant difference. Clearly, it is to our advantage to use a dependent sample whenever we can.

A second point can also be made regarding what we should do if we do not know for sure that the data is from a dependent sample. We should then go ahead and treat the data as if it were from an independent sample. If we can reject the null hypothesis under the assumption of independence, we can be confident that we would come to the same conclusion if the sample were dependent. Simply stated, the independent sample hypothesis test is a more conservative position.

When we are unable to verify that the data comes from a dependent sample and thus use the test statistic for an independent sample, we only take the chance of failing to reject a false null (a type II error). However, if we incorrectly assume that the data is from a dependent sample and use the test statistic for dependent samples, we take the chance of rejecting a true null hypothesis (a type I error).

Hot Spot #4 – Deciding Which Test Statistic To Use For Independent Samples

✷ *Mann Sections 10.1, 10.2 And 10.3*

One of the more confusing things about this chapter is deciding on which test statistic to use for independent samples. The sense of confusion can be reduced to a very low level by looking at the decision process using a tree diagram like we used in probability.

Our first decision is to choose between a large sample and small sample case. If both of the sample sizes are 30 or more, we immediately can find the value of the z test statistic and compare it to a critical value from the z table using the values of s_1^2 and s_2^2 for the values of σ_1^2 and σ_2^2. The first branch of the tree diagram can now be drawn. It is the upper branch of the tree diagram that follows in Figure 10–2 on the next page.

If either sample is less than 30, we need to use a t test statistic and there are two choices based on whether or not the population variances are equal:

If the population variances are assumed to be equal, we use the pooled standard deviation form of the t test. It is drawn in Figure 10–2 as the top part of the lower branch.

If the population variances are not equal, we then use the test statistic with a more conservative degrees of freedom calculated from the formula shown in the diagram below. The formula is in Figure 10–2 as the bottom part of the lower branch.

The completed tree diagram is presented below in Figure 10–2. It is important to keep a visual image of the tree diagram as we look at a hypothesis test for an independent sample. The mental picture makes it easier to go through the decision process.

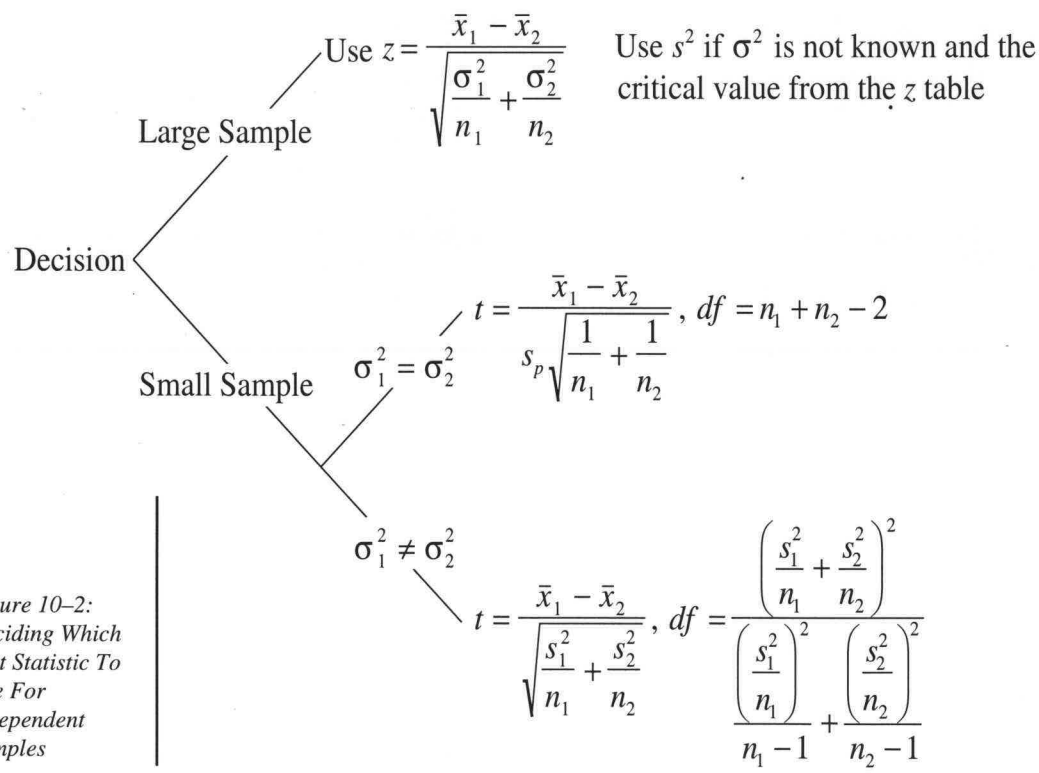

Figure 10–2: Deciding Which Test Statistic To Use For Independent Samples

CHAPTER 10 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the difference between an independent and dependent sample?

2. How does dependent sampling provide better control of confounding variables?

3. Why must we be concerned with the equality of population variances when we do hypothesis tests with two populations?

4. What is the test statistic for testing $\mu_1 - \mu_2$ for independent samples with equal variances?

5. Why do we use a smaller degree of freedom for a hypothesis tests about $\mu_1 - \mu_2$ when the population variances are not equal?

6. What test statistic do we use if the population variances are unequal and the size of the independent samples are small?

7. When can we use the z test statistic for a hypothesis test with independent samples from two populations?

8. When do we use the pooled estimate of p for a hypothesis test of two population proportions?

9. What is the difference between a controlled experiment and an observational experiment?

10. What is a double-blind randomized controlled experiment?